NEXARETAIL · LOYALTY APP REGISTRATION A/B TEST

Did the New Registration Page Win?

Prepared for Kavya Nair, Product Manager — 30-day experiment, 10,000 visitors, 50/50 Control vs. Treatment split

✓ SHIP THE NEW PAGE TO 100% OF USERS

The redesign produced a statistically significant, practically meaningful lift in registrations that holds directionally across every device and traffic channel tested.

THE NUMBERS

Control	Treatment	Absolute Lift	Relative Lift	P-value	95% CI (lift)
2.74% (137 / 5,004)	**3.60% (180 / 4,996)**	**+0.86 pp**	**+31.6%**	**0.0136**	**+0.18pp to +1.55pp**

Two-proportion z-test, two-sided (z = 2.47). P-value below the 0.05 threshold — reject the null hypothesis of no difference. The 95% confidence interval for the lift sits entirely above zero.

SEGMENT CHECK — DOES IT HOLD EVERYWHERE?

Segment	Control rate	Treatment rate	Rel. lift	P-value	Significant?
Mobile (65% of traffic)	2.44%	3.64%	+48.9%	0.0049	Yes
Desktop	3.27%	3.51%	+7.3%	0.730	No
Tablet	3.41%	3.66%	+7.4%	0.856	No
Direct	1.70%	3.39%	+99.5%	0.036	Yes
Google Ads	3.02%	3.04%	+0.5%	0.979	No
Instagram / WhatsApp / LinkedIn	2.02%–3.29%	3.56%–4.22%	+28% to +76%	> 0.05	No (small n)

Treatment leads in every device and every traffic channel — no segment reverses direction, so there's no Simpson's paradox distorting the overall result. Mobile (our largest segment) and Direct traffic are individually significant; other segments trend positive but are underpowered on their own at this sample size.

CAVEATS

- Single 30-day window — seasonal or promotional effects during March 2024 aren't ruled out.
- Google Ads traffic showed essentially flat conversion (+0.5%) — worth monitoring post-launch.
- Desktop, Tablet, and most individual channels trend positive but aren't independently significant; this reflects smaller segment sample sizes, not evidence the redesign underperforms there.

Analysis: two-proportion z-test on registration conversion (registered = 1), Control vs. Treatment. Full statistical workbook: NexaRetail_ABTest_ArchitVerma.ipynb — Prepared by Archit Verma